

# A Fundamental Tutorial on Fisher Information and the Cramér–Rao Lower Bound

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# 1 Introduction

In statistical signal processing, we are often faced with the following problem:

We observe a random variable (or signal) whose distribution depends on an unknown parameter, and we wish to estimate that parameter as accurately as possible.

Two central questions arise:

- How much **information** about the parameter is contained in the observations?
- What is the **best achievable accuracy** for any estimator?

Fisher Information answers the first question. The Cramér–Rao Lower Bound (CRLB) answers the second.

## 2 Basic Setup

Let  $X$  be a random variable with probability density function (pdf)

$$p(x; \theta),$$

where  $\theta \in \mathbb{R}$  is an unknown parameter.

We assume:

- $p(x; \theta)$  is differentiable in  $\theta$
- We can interchange differentiation and integration

We observe a sample  $X$ , and want to estimate  $\theta$ .

## 3 Estimators

An estimator is a function  $\hat{\theta}(X)$ .

### 3.1 Bias

The bias is:

$$\text{Bias}(\hat{\theta}) = \mathbb{E}[\hat{\theta}(X)] - \theta$$

An estimator is **unbiased** if:

$$\mathbb{E}[\hat{\theta}(X)] = \theta$$

### 3.2 Variance

$$\text{Var}(\hat{\theta}) = \mathbb{E}[(\hat{\theta} - \theta)^2]$$

We want estimators with low variance.

## 4 Likelihood and Log-Likelihood

Given observation  $x$ , the likelihood is:

$$L(\theta) = p(x; \theta)$$

Define the log-likelihood:

$$\ell(\theta) = \log p(x; \theta)$$

## 5 Score Function

### 5.1 Definition

The score function is:

$$u(X; \theta) = \frac{\partial}{\partial \theta} \log p(X; \theta)$$

## 5.2 Key Property: Zero Mean

We show:

$$\mathbb{E}[u(X; \theta)] = 0$$

**Proof:**

$$\mathbb{E}[u(X; \theta)] = \int \frac{\partial}{\partial \theta} \log p(x; \theta) p(x; \theta) dx$$

Using:

$$\begin{aligned} \frac{\partial}{\partial \theta} \log p &= \frac{1}{p} \frac{\partial p}{\partial \theta} \\ &= \int \frac{1}{p(x; \theta)} \frac{\partial p(x; \theta)}{\partial \theta} p(x; \theta) dx = \int \frac{\partial p(x; \theta)}{\partial \theta} dx \\ &= \frac{\partial}{\partial \theta} \int p(x; \theta) dx = \frac{\partial}{\partial \theta} (1) = 0 \end{aligned}$$

## 6 Fisher Information

### 6.1 Definition

The Fisher Information is:

$$\mathcal{I}(\theta) = \mathbb{E} \left[ \left( \frac{\partial}{\partial \theta} \log p(X; \theta) \right)^2 \right]$$

### 6.2 Alternative Form

We can also write:

$$\mathcal{I}(\theta) = -\mathbb{E} \left[ \frac{\partial^2}{\partial \theta^2} \log p(X; \theta) \right]$$

### 6.3 Proof of Equivalence

Start with:

$$\frac{\partial}{\partial \theta} p = p \cdot \frac{\partial}{\partial \theta} \log p$$

Differentiate again:

$$\frac{\partial^2 p}{\partial \theta^2} = p \left( \frac{\partial \log p}{\partial \theta} \right)^2 + p \cdot \frac{\partial^2 \log p}{\partial \theta^2}$$

Divide by  $p$ :

$$\frac{1}{p} \frac{\partial^2 p}{\partial \theta^2} = \left( \frac{\partial \log p}{\partial \theta} \right)^2 + \frac{\partial^2 \log p}{\partial \theta^2}$$

Take expectation:

$$\int \frac{\partial^2 p}{\partial \theta^2} dx = 0$$

Thus:

$$\mathbb{E} \left[ \left( \frac{\partial \log p}{\partial \theta} \right)^2 \right] = -\mathbb{E} \left[ \frac{\partial^2 \log p}{\partial \theta^2} \right]$$

The crucial step in the above proof is:

$$\int \frac{\partial^2 p}{\partial \theta^2} dx = 0$$

### 6.3.1 Why is the integral of second derivative zero?

#### 1. What does the expression mean?

We are interested in the expression:

$$\int \frac{\partial^2}{\partial \theta^2} p(x; \theta) dx$$

This involves:

- Integration over  $x$
- Differentiation twice with respect to  $\theta$

The key question is:

*Why does integrating the second derivative of the probability density function yield zero?*

#### 2. The core idea: total probability equals 1

For any valid probability density function:

$$\int p(x; \theta) dx = 1 \quad \text{for all } \theta$$

This is the fundamental starting point.

#### 3. Differentiate once

Differentiate both sides with respect to  $\theta$ :

$$\frac{\partial}{\partial \theta} \int p(x; \theta) dx = \frac{\partial}{\partial \theta} (1)$$

Now we use an important step: interchanging differentiation and integration:

$$\int \frac{\partial}{\partial \theta} p(x; \theta) dx = 0$$

#### 4. Differentiate again

Differentiate once more:

$$\frac{\partial}{\partial \theta} \left( \int \frac{\partial}{\partial \theta} p(x; \theta) dx \right) = 0$$

Again interchanging differentiation and integration gives:

$$\int \frac{\partial^2}{\partial \theta^2} p(x; \theta) dx = 0$$

## 5. Why is this allowed? (Regularity conditions)

This interchange is not automatically valid. It relies on **regularity conditions**, including:

- $p(x; \theta)$  is sufficiently smooth in  $\theta$
- derivatives are continuous
- integrals converge
- differentiation under the integral sign is valid

These are justified by results such as:

- Leibniz integral rule
- Dominated convergence theorem

## 6. Intuition

This result has a clear interpretation:

- Think of  $p(x; \theta)$  as a shape that moves with  $\theta$
- As  $\theta$  changes, the pdf may shift or deform
- However, the total area must always remain 1

Therefore:

- The first derivative integrates to zero  $\Rightarrow$  no net change in total probability
- The second derivative integrates to zero  $\Rightarrow$  the rate of change of that net change is also zero

**The total probability mass is “locked” at 1**

## 7. Why this matters for Fisher Information

This identity is crucial for proving the equivalence:

$$\mathcal{I}(\theta) = \mathbb{E} \left[ \left( \frac{\partial}{\partial \theta} \log p(X; \theta) \right)^2 \right] = -\mathbb{E} \left[ \frac{\partial^2}{\partial \theta^2} \log p(X; \theta) \right]$$

Without:

$$\int \frac{\partial^2}{\partial \theta^2} p(x; \theta) dx = 0$$

this equivalence does not hold.

## 8. A subtle but important caveat

This result can fail in certain cases, particularly when:

- the support of  $p(x; \theta)$  depends on  $\theta$
- boundary terms appear during differentiation

### Example: Uniform distribution

Consider:

$$X \sim \text{Uniform}(0, \theta)$$

Here:

- the upper limit of integration depends on  $\theta$
- differentiation introduces boundary terms

In such cases:

$$\int \frac{\partial^2}{\partial \theta^2} p(x; \theta) dx \neq 0$$

This highlights the importance of the assumptions underlying results such as the Cramér–Rao Lower Bound.

## 7 Interpretation of Fisher Information

Fisher Information measures:

**How sensitive the probability distribution is to changes in the parameter.**

- If small changes in  $\theta$  cause large changes in  $p(x; \theta)$ , information is high
- If  $p(x; \theta)$  barely changes, information is low

## 8 Multiple Observations

For i.i.d. samples  $X_1, \dots, X_N$ :

$$\mathcal{I}_N(\theta) = N\mathcal{I}_1(\theta)$$

**Interpretation:** Information adds linearly.

## 9 Cramér–Rao Lower Bound (CRLB)

### 9.1 Statement

For any unbiased estimator:

$$\text{Var}(\hat{\theta}) \geq \frac{1}{\mathcal{I}(\theta)}$$

## 10 Proof of the CRLB

Define:

$$u(X; \theta) = \frac{\partial}{\partial \theta} \log p(X; \theta)$$

We use:

$$\mathbb{E}[\hat{\theta}] = \theta$$

Differentiate:

$$\frac{d}{d\theta}\mathbb{E}[\hat{\theta}] = 1$$

Move derivative inside (using definition of expectation operator):

$$\int \hat{\theta}(x) \frac{\partial p(x; \theta)}{\partial \theta} dx = 1$$

Rewrite:

$$\int \hat{\theta}(x) p(x; \theta) \frac{\partial \log p(x; \theta)}{\partial \theta} dx = 1$$

Thus:

$$\mathbb{E}[\hat{\theta} u(X; \theta)] = 1$$

Write:

$$\hat{\theta} = (\hat{\theta} - \theta) + \theta$$

Then:

$$\mathbb{E}[\hat{\theta} u] = \mathbb{E}[(\hat{\theta} - \theta)u + \theta u] = \mathbb{E}[(\hat{\theta} - \theta)u] + \theta \mathbb{E}[u]$$

For regular models:

$$\mathbb{E}[u(X; \theta)] = 0$$

Thus:

$$\mathbb{E}[\hat{\theta} u] = \mathbb{E}[(\hat{\theta} - \theta)u]$$

Hence:

$$\mathbb{E}[(\hat{\theta} - \theta)u(X; \theta)] = 1$$

Now apply Cauchy–Schwarz:

$$\left( \mathbb{E}[(\hat{\theta} - \theta)u] \right)^2 \leq \mathbb{E}[(\hat{\theta} - \theta)^2] \cdot \mathbb{E}[u^2]$$

Thus:

$$1^2 \leq \text{Var}(\hat{\theta}) \cdot \mathcal{I}(\theta)$$

$$\Rightarrow \text{Var}(\hat{\theta}) \geq \frac{1}{\mathcal{I}(\theta)}$$

## 11 Geometric Interpretation

Think of:

- $\hat{\theta} - \theta$  as an error vector
- $u(X; \theta)$  as a direction of sensitivity

CRLB arises from projecting estimation error onto the score direction.



## 11.1 Hilbert Space and Orthogonality Interpretation of the CRLB

### 11.1.1 The Hilbert space of random variables

Consider the space of random variables with finite second moment:

$$\mathcal{H} = \{Z : \mathbb{E}[Z^2] < \infty\}$$

Define an inner product:

$$\langle X, Y \rangle = \mathbb{E}[XY]$$

Then:

- $\mathcal{H}$  is a Hilbert space
- Norm:

$$\|X\|^2 = \mathbb{E}[X^2]$$

### 11.1.2 Key quantities as vectors

Define:

- Estimation error:

$$e(X) = \hat{\theta}(X) - \theta$$

- Score function:

$$u(X; \theta) = \frac{\partial}{\partial \theta} \log p(X; \theta)$$

Both  $e$  and  $u$  are elements of  $\mathcal{H}$ .

### 11.1.3 A crucial identity

From the CRLB derivation:

$$\mathbb{E}[(\hat{\theta} - \theta)u(X; \theta)] = 1$$

In Hilbert space notation:

$$\langle e, u \rangle = 1$$

### 11.1.4 Applying Cauchy–Schwarz

The Cauchy–Schwarz inequality gives:

$$|\langle e, u \rangle|^2 \leq \|e\|^2 \cdot \|u\|^2$$

Substitute:

$$1^2 \leq \mathbb{E}[e^2] \cdot \mathbb{E}[u^2]$$

Thus:

$$\text{Var}(\hat{\theta}) \geq \frac{1}{\mathcal{I}(\theta)}$$

### 11.1.5 Geometric interpretation

Think of:

- $e$  as a vector (estimation error)
- $u$  as a direction of sensitivity in the likelihood

The quantity:

$$\langle e, u \rangle$$

is the projection of the error onto the score direction.

The constraint:

$$\langle e, u \rangle = 1$$

means the error must have a fixed component along  $u$ .

### 11.1.6 Why variance cannot be arbitrarily small

We want to minimize:

$$\|e\|^2 = \mathbb{E}[e^2]$$

subject to:

$$\langle e, u \rangle = 1$$

This is a constrained minimization problem in a Hilbert space.

The solution is:

$$e = \lambda u$$

Substitute into constraint:

$$\langle \lambda u, u \rangle = \lambda \|u\|^2 = 1$$

$$\lambda = \frac{1}{\|u\|^2} = \frac{1}{\mathcal{I}(\theta)}$$

Thus:

$$\|e\|^2 = \lambda^2 \|u\|^2 = \frac{1}{\mathcal{I}(\theta)}$$

### 11.1.7 Interpretation of efficient estimators

The lower bound is achieved when:

$$e(X) = \frac{1}{\mathcal{I}(\theta)} u(X; \theta)$$

That is:

**The estimation error lies entirely in the direction of the score function.**

### 11.1.8 Orthogonality viewpoint

More generally, decompose:

$$e = \underbrace{\alpha u}_{\text{aligned with score}} + \underbrace{e_{\perp}}_{\text{orthogonal to score}}$$

with:

$$\langle e_{\perp}, u \rangle = 0$$

Then:

$$\|e\|^2 = \alpha^2 \|u\|^2 + \|e_{\perp}\|^2$$

The constraint forces:

$$\alpha = \frac{1}{\|u\|^2}$$

Thus:

$$\text{Var}(\hat{\theta}) = \frac{1}{\mathcal{I}(\theta)} + \|e_{\perp}\|^2$$

### 11.1.9 Final insight

- The CRLB corresponds to the case where  $e_{\perp} = 0$
- Any additional orthogonal component increases variance

**Extra variance comes from error components orthogonal to the score function.**

## 12 Efficient Estimators

An estimator is **efficient** if:

$$\text{Var}(\hat{\theta}) = \frac{1}{\mathcal{I}(\theta)}$$

This happens if:

$$\hat{\theta} - \theta = c \cdot u(X; \theta)$$

## 13 Example: Gaussian Mean Estimation

Let:

$$X \sim \mathcal{N}(\mu, \sigma^2)$$

### 13.1 Compute Fisher Information

$$\log p = -\frac{(x - \mu)^2}{2\sigma^2} + \text{const}$$

$$\frac{\partial}{\partial \mu} \log p = \frac{x - \mu}{\sigma^2}$$

$$\mathcal{I}(\mu) = \mathbb{E} \left[ \left( \frac{x - \mu}{\sigma^2} \right)^2 \right] = \frac{1}{\sigma^2}$$

## 13.2 CRLB

$$\text{Var}(\hat{\mu}) \geq \frac{\sigma^2}{N}$$

Sample mean achieves this bound.

## 14 Vector Parameters

For  $\boldsymbol{\theta} \in \mathbb{R}^d$ :

$$\mathbf{I}(\boldsymbol{\theta}) = \mathbb{E} [(\nabla \log p)(\nabla \log p)^T]$$

CRLB:

$$\text{Cov}(\hat{\boldsymbol{\theta}}) \succeq \mathbf{I}^{-1}(\boldsymbol{\theta})$$

## 15 Connection to Signal Processing

In signal processing:

- $\theta$  could be delay, frequency, amplitude
- Fisher Information depends on waveform structure and noise

High bandwidth  $\Rightarrow$  higher Fisher Information.

## 16 Summary

- Fisher Information measures sensitivity of likelihood to parameter
- CRLB gives a fundamental lower bound on estimator variance
- Information adds with independent observations
- Efficient estimators achieve the bound

## 17 Exercises

1. Derive Fisher Information for variance of Gaussian
2. Show exponential distribution FI
3. Extend CRLB to biased estimators

## A Regularity Conditions and Non-Regular Models

### A.1 When does $\int \frac{\partial^2}{\partial \theta^2} p(x; \theta) dx = 0$ hold?

The identity

$$\int \frac{\partial^2}{\partial \theta^2} p(x; \theta) dx = 0$$

is not universally valid. It holds under specific **regularity conditions**, which are often implicitly assumed in statistical signal processing.

#### A.1.1 Key condition: Parameter-independent support

A crucial requirement is that the **support of the distribution does not depend on  $\theta$** .

That is, there exists a fixed set  $\mathcal{X}$  such that:

$$p(x; \theta) > 0 \quad \text{only for } x \in \mathcal{X}, \quad \text{independent of } \theta$$

**Examples where it *does* work**

- Gaussian:

$$X \sim \mathcal{N}(\mu, \sigma^2), \quad x \in (-\infty, \infty)$$

- Exponential:

$$X \sim \text{Exp}(\lambda), \quad x \in [0, \infty)$$

- Laplace, Gamma, etc.

In all these cases:

- The limits of integration do not depend on  $\theta$
- Differentiation can safely pass through the integral

#### A.1.2 When it fails: Parameter-dependent support

Consider:

$$X \sim \text{Uniform}(0, \theta)$$

The pdf is:

$$p(x; \theta) = \begin{cases} \frac{1}{\theta}, & 0 \leq x \leq \theta \\ 0, & \text{otherwise} \end{cases}$$

Here, the support depends on  $\theta$ :

$$\mathcal{X}(\theta) = [0, \theta]$$

## Why the derivation breaks

Previously, we wrote:

$$\frac{\partial}{\partial \theta} \int p(x; \theta) dx = \int \frac{\partial}{\partial \theta} p(x; \theta) dx$$

This is no longer valid because the limits depend on  $\theta$ .

We must instead use the **Leibniz rule**:

$$\frac{\partial}{\partial \theta} \int_0^\theta p(x; \theta) dx = \int_0^\theta \frac{\partial}{\partial \theta} p(x; \theta) dx + p(\theta; \theta) \cdot \frac{d\theta}{d\theta}$$

## Compute explicitly

Since:

$$\int_0^\theta \frac{1}{\theta} dx = 1$$

Differentiate:

$$\frac{\partial}{\partial \theta} \int_0^\theta \frac{1}{\theta} dx = 0$$

Now compute RHS:

$$\begin{aligned} & \int_0^\theta \frac{\partial}{\partial \theta} \left( \frac{1}{\theta} \right) dx + \frac{1}{\theta} \\ &= \int_0^\theta \left( -\frac{1}{\theta^2} \right) dx + \frac{1}{\theta} = -\frac{\theta}{\theta^2} + \frac{1}{\theta} = -\frac{1}{\theta} + \frac{1}{\theta} = 0 \end{aligned}$$

## Key insight

The boundary term cancels the interior derivative term.

**This is exactly what is missing in the standard Fisher Information derivation.**

### A.1.3 Implication for Fisher Information

For regular models:

$$\mathcal{I}(\theta) = -\mathbb{E} \left[ \frac{\partial^2}{\partial \theta^2} \log p(X; \theta) \right]$$

For non-regular models (like Uniform):

- This identity fails
- The Fisher Information must be computed using:

$$\mathcal{I}(\theta) = \mathbb{E} \left[ \left( \frac{\partial}{\partial \theta} \log p(X; \theta) \right)^2 \right]$$

#### A.1.4 Even deeper issue: the score function

For the Uniform case:

$$\log p(x; \theta) = -\log \theta$$

$$\frac{\partial}{\partial \theta} \log p = -\frac{1}{\theta}$$

This does *not depend on x*, so:

$$\mathbb{E}[u(X; \theta)] = -\frac{1}{\theta} \neq 0$$

This violates a fundamental assumption:

$$\mathbb{E}[u(X; \theta)] = 0$$

**Conclusion: the model is non-regular.**

#### A.1.5 What does this mean practically?

- The standard CRLB does *not apply*
- The MLE behaves differently (for Uniform,  $\hat{\theta} = \max X_i$ )
- Estimation performance must be analyzed using alternative tools

#### A.1.6 Big picture intuition

- Regular models: parameter changes *shape* of pdf
- Uniform case: parameter changes *support boundary*

**Fisher Information assumes smooth changes in likelihood — not moving boundaries.**

## B Detailed Solutions to Exercises

### B.1 Exercise 1: Fisher Information for the Variance of a Gaussian

Suppose

$$X \sim \mathcal{N}(\mu, \sigma^2),$$

where  $\mu$  is known and the unknown parameter is

$$\theta = \sigma^2.$$

The density is

$$p(x; \theta) = \frac{1}{\sqrt{2\pi\theta}} \exp\left(-\frac{(x - \mu)^2}{2\theta}\right).$$

The log-likelihood for one observation is

$$\ell(\theta) = \log p(x; \theta) = -\frac{1}{2} \log(2\pi\theta) - \frac{(x - \mu)^2}{2\theta}.$$

Differentiate with respect to  $\theta$ :

$$\frac{\partial \ell}{\partial \theta} = -\frac{1}{2\theta} + \frac{(x - \mu)^2}{2\theta^2}.$$

This is the score function:

$$u(X; \theta) = -\frac{1}{2\theta} + \frac{(X - \mu)^2}{2\theta^2}.$$

The Fisher Information is

$$\mathcal{I}(\theta) = \mathbb{E}[u(X; \theta)^2].$$

It is easier to use the second-derivative form:

$$\mathcal{I}(\theta) = -\mathbb{E}\left[\frac{\partial^2 \ell}{\partial \theta^2}\right].$$

Now,

$$\frac{\partial^2 \ell}{\partial \theta^2} = \frac{1}{2\theta^2} - \frac{(x - \mu)^2}{\theta^3}.$$

Therefore,

$$\mathcal{I}(\theta) = -\mathbb{E}\left[\frac{1}{2\theta^2} - \frac{(X - \mu)^2}{\theta^3}\right].$$

Since

$$\mathbb{E}[(X - \mu)^2] = \theta,$$

we obtain

$$\mathcal{I}(\theta) = -\left[\frac{1}{2\theta^2} - \frac{\theta}{\theta^3}\right].$$

Thus,

$$\mathcal{I}(\theta) = -\left[\frac{1}{2\theta^2} - \frac{1}{\theta^2}\right] = \frac{1}{2\theta^2}.$$

Hence, for one observation,

$$\boxed{\mathcal{I}(\theta) = \frac{1}{2\theta^2}}$$



Since  $\theta = \sigma^2$ , this may also be written as

$$\mathcal{I}(\sigma^2) = \frac{1}{2\sigma^4}$$

For  $N$  independent observations,

$$\mathcal{I}_N(\sigma^2) = \frac{N}{2\sigma^4}$$

Therefore, the Cramér–Rao lower bound for any unbiased estimator of  $\sigma^2$  is

$$\text{Var}(\widehat{\sigma^2}) \geq \frac{2\sigma^4}{N}$$

## B.2 Exercise 2: Fisher Information for the Exponential Distribution

Suppose

$$X \sim \text{Exp}(\lambda),$$

where  $\lambda > 0$  is the unknown rate parameter.

The density is

$$p(x; \lambda) = \lambda e^{-\lambda x}, \quad x \geq 0.$$

The log-likelihood for one observation is

$$\ell(\lambda) = \log p(x; \lambda) = \log \lambda - \lambda x.$$

Differentiate:

$$\frac{\partial \ell}{\partial \lambda} = \frac{1}{\lambda} - x.$$

Thus the score function is

$$u(X; \lambda) = \frac{1}{\lambda} - X.$$

The Fisher Information is

$$\mathcal{I}(\lambda) = \mathbb{E} \left[ \left( \frac{1}{\lambda} - X \right)^2 \right].$$

Since for an exponential random variable,

$$\mathbb{E}[X] = \frac{1}{\lambda}, \quad \text{Var}(X) = \frac{1}{\lambda^2},$$

we have

$$\mathcal{I}(\lambda) = \text{Var}(X) = \frac{1}{\lambda^2}.$$

Therefore, for one observation,

$$\mathcal{I}(\lambda) = \frac{1}{\lambda^2}$$

For  $N$  independent observations,

$$\mathcal{I}_N(\lambda) = \frac{N}{\lambda^2}$$

Hence the CRLB for any unbiased estimator of  $\lambda$  is

$$\text{Var}(\hat{\lambda}) \geq \frac{\lambda^2}{N}$$

### B.2.1 Alternative parameterisation using the mean

Sometimes the exponential distribution is parameterised by its mean

$$\theta = \frac{1}{\lambda}.$$

Then the density is

$$p(x; \theta) = \frac{1}{\theta} e^{-x/\theta}, \quad x \geq 0.$$

The log-likelihood is

$$\ell(\theta) = -\log \theta - \frac{x}{\theta}.$$

Differentiate:

$$\frac{\partial \ell}{\partial \theta} = -\frac{1}{\theta} + \frac{x}{\theta^2}.$$

The second derivative is

$$\frac{\partial^2 \ell}{\partial \theta^2} = \frac{1}{\theta^2} - \frac{2x}{\theta^3}.$$

Using

$$\mathbb{E}[X] = \theta,$$

we get

$$\mathcal{I}(\theta) = -\mathbb{E} \left[ \frac{1}{\theta^2} - \frac{2X}{\theta^3} \right].$$

Therefore,

$$\mathcal{I}(\theta) = - \left[ \frac{1}{\theta^2} - \frac{2\theta}{\theta^3} \right] = - \left[ \frac{1}{\theta^2} - \frac{2}{\theta^2} \right] = \frac{1}{\theta^2}.$$

Thus,

$$\boxed{\mathcal{I}(\theta) = \frac{1}{\theta^2}}$$

For  $N$  independent observations,

$$\boxed{\mathcal{I}_N(\theta) = \frac{N}{\theta^2}}$$

### B.3 Exercise 3: Extension of the CRLB to Biased Estimators

The standard CRLB assumes that the estimator is unbiased:

$$\mathbb{E}[\hat{\theta}] = \theta.$$

However, many useful estimators are biased. We therefore consider the more general case. Let the bias be

$$b(\theta) = \mathbb{E}[\hat{\theta}] - \theta.$$

Thus,

$$\mathbb{E}[\hat{\theta}] = \theta + b(\theta).$$

Differentiate both sides with respect to  $\theta$ :

$$\frac{d}{d\theta} \mathbb{E}[\hat{\theta}] = 1 + b'(\theta).$$

Now write the expectation as an integral:

$$\mathbb{E}[\hat{\theta}] = \int \hat{\theta}(x)p(x; \theta) dx.$$

Assuming the usual regularity conditions, we can move the derivative inside the integral:

$$\frac{d}{d\theta}\mathbb{E}[\hat{\theta}] = \int \hat{\theta}(x) \frac{\partial}{\partial \theta} p(x; \theta) dx.$$

Using

$$\frac{\partial}{\partial \theta} p(x; \theta) = p(x; \theta) \frac{\partial}{\partial \theta} \log p(x; \theta),$$

we obtain

$$1 + b'(\theta) = \int \hat{\theta}(x)p(x; \theta) \frac{\partial}{\partial \theta} \log p(x; \theta) dx.$$

Therefore,

$$1 + b'(\theta) = \mathbb{E} \left[ \hat{\theta}(X) u(X; \theta) \right],$$

where

$$u(X; \theta) = \frac{\partial}{\partial \theta} \log p(X; \theta)$$

is the score function.

Since

$$\mathbb{E}[u(X; \theta)] = 0,$$

we may subtract  $\theta$  inside the expectation:

$$\mathbb{E} \left[ \hat{\theta} u \right] = \mathbb{E} \left[ (\hat{\theta} - \theta) u \right].$$

Therefore,

$$\mathbb{E} \left[ (\hat{\theta} - \theta) u \right] = 1 + b'(\theta).$$

Now apply Cauchy–Schwarz:

$$\left( \mathbb{E} \left[ (\hat{\theta} - \theta) u \right] \right)^2 \leq \mathbb{E} \left[ (\hat{\theta} - \theta)^2 \right] \mathbb{E} \left[ u^2 \right].$$

Thus,

$$(1 + b'(\theta))^2 \leq \mathbb{E} \left[ (\hat{\theta} - \theta)^2 \right] \mathcal{I}(\theta).$$

Hence,

$$\mathbb{E} \left[ (\hat{\theta} - \theta)^2 \right] \geq \frac{(1 + b'(\theta))^2}{\mathcal{I}(\theta)}.$$

The left-hand side is the mean squared error:

$$\text{MSE}(\hat{\theta}) = \mathbb{E} \left[ (\hat{\theta} - \theta)^2 \right].$$

Therefore, the biased CRLB for mean squared error is

$$\boxed{\text{MSE}(\hat{\theta}) \geq \frac{(1 + b'(\theta))^2}{\mathcal{I}(\theta)}}$$

Now recall the bias–variance decomposition:

$$\text{MSE}(\hat{\theta}) = \text{Var}(\hat{\theta}) + b(\theta)^2.$$

Therefore,

$$\text{Var}(\hat{\theta}) + b(\theta)^2 \geq \frac{(1 + b'(\theta))^2}{\mathcal{I}(\theta)}.$$

Rearranging gives the biased CRLB for the variance:

$$\boxed{\text{Var}(\hat{\theta}) \geq \frac{(1 + b'(\theta))^2}{\mathcal{I}(\theta)} - b(\theta)^2}$$

### B.3.1 Understanding the bias–variance decomposition

The mean squared error (MSE) is defined as:

$$\text{MSE}(\hat{\theta}) = \mathbb{E} \left[ (\hat{\theta} - \theta)^2 \right].$$

This measures the average squared distance between the estimator and the true parameter.

#### Derivation

Let

$$m(\theta) = \mathbb{E}[\hat{\theta}]$$

be the mean of the estimator. Then we write:

$$\hat{\theta} - \theta = (\hat{\theta} - m(\theta)) + (m(\theta) - \theta).$$

Define the bias:

$$b(\theta) = m(\theta) - \theta.$$

Thus:

$$\hat{\theta} - \theta = (\hat{\theta} - \mathbb{E}[\hat{\theta}]) + b(\theta).$$

Now square both sides:

$$(\hat{\theta} - \theta)^2 = (\hat{\theta} - \mathbb{E}[\hat{\theta}])^2 + 2b(\theta) (\hat{\theta} - \mathbb{E}[\hat{\theta}]) + b(\theta)^2.$$

Take expectations:

$$\mathbb{E} \left[ (\hat{\theta} - \theta)^2 \right] = \mathbb{E} \left[ (\hat{\theta} - \mathbb{E}[\hat{\theta}])^2 \right] + 2b(\theta) \mathbb{E} \left[ \hat{\theta} - \mathbb{E}[\hat{\theta}] \right] + b(\theta)^2.$$

The middle term vanishes since:

$$\mathbb{E} \left[ \hat{\theta} - \mathbb{E}[\hat{\theta}] \right] = 0.$$

Therefore:

$$\text{MSE}(\hat{\theta}) = \text{Var}(\hat{\theta}) + b(\theta)^2.$$

## Interpretation

The decomposition separates the total estimation error into two fundamentally different components:

- **Variance:**

$$\text{Var}(\hat{\theta}) = \mathbb{E} \left[ \left( \hat{\theta} - \mathbb{E}[\hat{\theta}] \right)^2 \right]$$

This measures the *random spread* of the estimator due to noise.

- **Squared bias:**

$$b(\theta)^2 = \left( \mathbb{E}[\hat{\theta}] - \theta \right)^2$$

This measures the *systematic offset* of the estimator from the true value.

## Geometric intuition

One may think of  $\hat{\theta}$  as producing a cloud of estimates:

- The variance measures how wide the cloud is (random uncertainty)
- The bias measures how far the centre of the cloud is from the true value

Thus, the MSE measures:

**the total squared distance from the estimates to the true parameter**

which naturally splits into spread plus offset.

## Why this matters

Even if an estimator has small variance, it may still perform poorly if it is biased. Conversely, a slightly biased estimator may have much smaller variance and thus lower overall MSE.

This decomposition is therefore fundamental when analysing estimator performance, especially when extending the Cramér–Rao bound to biased estimators.

## Connection to signal processing

In signal processing applications (e.g. delay estimation):

- Variance arises from noise causing fluctuations in the detected parameter
- Bias arises from systematic effects such as filtering delays, windowing, or model mismatch

Thus, even a stable estimator may produce incorrect results if it is biased, and the MSE provides the correct measure of total error.

### B.3.2 Check: recovery of the unbiased CRLB

If the estimator is unbiased, then

$$b(\theta) = 0, \quad b'(\theta) = 0.$$

The biased CRLB becomes

$$\text{Var}(\hat{\theta}) \geq \frac{1}{\mathcal{I}(\theta)},$$

which is exactly the ordinary Cramér–Rao lower bound.

### B.3.3 Interpretation

For a biased estimator, the lower bound depends not only on the Fisher Information

$$\mathcal{I}(\theta),$$

but also on how the bias changes with the parameter:

$$b'(\theta).$$

This is important because bias is not always bad. A biased estimator may have lower variance than any unbiased estimator, but the relevant performance measure is often the mean squared error:

$$\text{MSE}(\hat{\theta}) = \text{Var}(\hat{\theta}) + b(\theta)^2.$$

Thus, the biased CRLB is especially useful when studying bias–variance tradeoffs.